Aarhus University

Computer Science

Numerical Linear Algebra

Handin 8

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Consider the matrix

$$A = \begin{bmatrix} 3 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 1 & 1 \end{bmatrix}.$$

(a) Characteristic Polynomial

Determine the characteristic polynomial $\det(A - \lambda I_3)$ of A.

$$A - \lambda I_3 = \begin{bmatrix} 3 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 1 & 1 \end{bmatrix} - \begin{bmatrix} \lambda & 0 & 0 \\ 0 & \lambda & 0 \\ 0 & 0 & \lambda \end{bmatrix} = \begin{bmatrix} 3-\lambda & 0 & 0 \\ 0 & 1-\lambda & 1 \\ 0 & 1 & 1-\lambda \end{bmatrix} \tag{1}$$

$$\det(A-\lambda I_3) = (3-\lambda)\cdot\begin{bmatrix} 1-\lambda & 1 \\ 1 & 1-\lambda \end{bmatrix} - 0\cdot\begin{bmatrix} 0 & 1 \\ 0 & 1-\lambda \end{bmatrix} + 0\cdot\begin{bmatrix} 0 & 1-\lambda \\ 0 & 1 \end{bmatrix} = (3-\lambda)\cdot((1-\lambda)\cdot(1-\lambda)-(1\cdot 1))$$

Definition 21.12 [2]

$$(3-\lambda)\cdot((1-\lambda)\cdot(1-\lambda)-(1\cdot 1)) = (3-\lambda)\cdot(\lambda^2 - 2\lambda)$$

Conclusion:

$$\det(A - \lambda I_3) = (3-\lambda)\cdot(\lambda^2 - 2\lambda)$$

(b-c) Some Eigenvalues

Show that 0 and 3 are eigenvalues of A and other values are welcome too.

The eigenvalues λ can be found by finding the solutions to the characteristic polynomial from (a), since any λ that results in the polynomials being 0 is an eigenvalue for A.
This means solving:

$$(3-\lambda)\cdot(\lambda^2 - 2\lambda) = 0$$

(With a hint from MatLab) this can formulated to finding a solution to:

$$3 - \lambda = 0 \quad \text{or} \quad \lambda^2 - 2\lambda = 0$$

trivially a solution is $\lambda = 3$, making 3 an eigenvalue for A:

$$3 - \lambda = 0 = 3 - 3$$

For the other one $\lambda^2 - 2\lambda$, the quadratic formula can be used
to find the roots for $1 \cdot \lambda^2 - 2 \cdot \lambda + 0 = 0$.

$$\lambda = \frac{-(-2) \pm \sqrt{(-2)^2 - 4 \cdot 1 \cdot 0}}{2 \cdot 1} = \frac{2 \pm \sqrt{4}}{2} = \frac{2 \pm 2}{2}$$

This leaves two solutions

$$\frac{2+2}{2} = 2 \quad \text{and} \quad \frac{2-2}{2} = 0$$

which means the the eigenvalues for A is $[3, 2, 0]$

(d) Basis using Eigenvectors

Determine a basis of $\mathbb{R}^3$ consisting of eigenvectors of A.

Proposition 21.5 [2], says that if the eigenvalues of A are unique and there is a set of eigenvectores for A, then the eigenvectores constitues a basis for A.
Eigenvectores can be found by solving for a v in $(A - \lambda I_3)v = 0$ for each of the eigenvalues $\lambda \in [3, 2, 0]$

From equation(1) $(A - \lambda I_3)$ is already expressed.
For $\lambda = 3$

$$\begin{bmatrix} 3-3 & 0 & 0 \\ 0 & 1-3 & 1 \\ 0 & 1 & 1-3 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & -2 & 1 \\ 0 & 1 & -2 \end{bmatrix} = (A - 3I_3)$$

add some $v \in \mathbb{R}^3 \neq 0$ (Does this notation even make sense?)

$$(A - 3I_3)v = \begin{bmatrix} 0 & 0 & 0 \\ 0 & -2 & 1 \\ 0 & 1 & -2 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

Note, there are many solutions for v since x can litereally be any value, but a basis needs to conist of linear independant vectors, which means only a single vector per eigenvalue is needed, hence the simplest vector that can be found is the best.

$$(A - 3I_3)v = \begin{bmatrix} 0 & 0 & 0 \\ 0 & -2 & 1 \\ 0 & 1 & -2 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

The vector $\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$ is an eigenvector for A.

For $\lambda = 2$

$$(A - 2I_3) = \begin{bmatrix} 3-2 & 0 & 0 \\ 0 & 1-2 & 1 \\ 0 & 1 & 1-2 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 1 \\ 0 & 1 & -1 \end{bmatrix}$$

$$(A - 2I_3)v = \begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 1 \\ 0 & 1 & -1 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

For $\lambda = 0$

$$(A - 1I_3) = \begin{bmatrix} 3-0 & 0 & 0 \\ 0 & 1-0 & 1 \\ 0 & 1 & 1-0 \end{bmatrix} = \begin{bmatrix} 3 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 1 & 1 \end{bmatrix}$$

$$(A - 1I_3)v = \begin{bmatrix} 3 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ -1 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

Now for the eigenvalues $[3, 2, 0]$ we have the following eigenvectores $\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ -1 \\ 1 \end{bmatrix}$ Which constitutes the basis for A

(e) Panic?!

Write $A = V\Lambda V^{-1}$ for suitable V and diagonal Λ.
Following Proposition 21.6 [2] with the resulting basis for A from (d) and the eigenvalues from (b-c).

$$V = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & -1 \\ 0 & 1 & 1 \end{bmatrix} \qquad \Lambda = \begin{bmatrix} 3 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 0 \end{bmatrix} \qquad V^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \frac{1}{2} & \frac{1}{2} \\ 0 & -\frac{1}{2} & \frac{1}{2} \end{bmatrix}$$

The inverse of V was calculated using python and includ a sanity check for $A = V\Lambda V^{-1}$

```
# Defining the Diagonalization of A
V = np.array([[1,0,0],
              [0,1,-1],
              [0,1,1]])
Vinv = np.linalg.inv(V)
Lambda = np.array([[3,0,0],
                   [0,2,0],
                   [0,0,0]])
# Sanity check for A
A = V @ Lambda @ Vinv
print(A)
#Output -->
[[3. 0. 0.]
 [0. 1. 1.]
 [0. 1. 1.]]
```

hence A is Diagonalizable.

(f) Witness me! - Said Donald Duck Calmly

Use item (e) to determine A^k for all k.

Item (e) is the factorization of $A = V\Lambda V^{-1}$, refering to Theorem 6.3.2 [1] remark 4:
If A is diagonalizable, then A can be factored into a product $V\Lambda V^{-1}$
it then follows from remark 4 that

$$A^2 = (V\Lambda V^{-1})(V\Lambda V^{-1}) = V\Lambda^2 V^{-1}$$

This leaves $A^k = V\Lambda^k V^{-1}$ which can be refined further by focusing on the right side.

$$V\Lambda^k V^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & -1 \\ 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} 3^k & 0 & 0 \\ 0 & 2^k & 0 \\ 0 & 0 & 0^k \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & \frac{1}{2} & \frac{1}{2} \\ 0 & -\frac{1}{2} & \frac{1}{2} \end{bmatrix}$$

Let's split this Mammoth of matrix multiplicaiton

$$V\Lambda^k = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & -1 \\ 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} 3^k & 0 & 0 \\ 0 & 2^k & 0 \\ 0 & 0 & 0^k \end{bmatrix} = \begin{bmatrix} 3^k & 0 & 0 \\ 0 & 2^k & 0 \\ 0 & 2^k & 0 \end{bmatrix}$$

Then multiply by V^{-1}, remember $\frac{2^k}{2} = 2^{k-1}$

$$V\Lambda^k V^{-1} = \begin{bmatrix} 3^k & 0 & 0 \\ 0 & 2^k & 0 \\ 0 & 2^k & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & \frac{1}{2} & \frac{1}{2} \\ 0 & -\frac{1}{2} & \frac{1}{2} \end{bmatrix} = \begin{bmatrix} 3^k & 0 & 0 \\ 0 & 2^{k-1} & 2^{k-1} \\ 0 & 2^{k-1} & 2^{k-1} \end{bmatrix}$$

In conclusion A^k can be expressed/determined using item (e) as such

$$A^k = \begin{bmatrix} 3^k & 0 & 0 \\ 0 & 2^{k-1} & 2^{k-1} \\ 0 & 2^{k-1} & 2^{k-1} \end{bmatrix}$$

Appendix

Only code used is explicitely shown when applied.

References

[1] Steven J. Leaon. *Numerical Linear Algebra with Applications, Ninth Edition.* Pearson Education, Inc, 2015.

[2] Andrew Swan. *Numerisk Linær Algebra.* Institut for Matematik, Aarhus Universitet, 2025.